

ANDREW LIN

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EDUCATION

University of Waterloo

Bachelor of Computer Science, Co-op, GPA: 3.93

- Dean's Honours List, 2023 – 2025
- Relevant Courses: Data Structures and Algorithms, Logic and Computation, Object-Oriented Software Development

Waterloo, ON

Expected Apr. 2027

EXPERIENCE

Software Engineer Intern

Geotab

- Building data ingestion services and pipelines for 200K commercial vehicles using **C#**, **GCP**, **Kubernetes**, and **Terraform**

Jan. 2025 – Present

Waterloo, ON

Software Engineer Intern

TD Bank

- Led development of an Enterprise Innovation team project delivering near real-time translation between clients and bank tellers, built with **FastAPI**, **React**, and deployed on **GCP Compute Engine** and **Heroku**
- Achieved a 6% increase in translation performance measured by string-based and ML-based metrics (eg. BLEU, COMET) by evaluating and comparing the quality of four different translation models across 38 language pairs
- Developed FastAPI endpoints to evaluate translation quality and handle audio playback through **WebSocket** connections
- Built a single-device conversation feature with automatic speaker differentiation using **Whisper**'s language detection

May 2024 – Aug. 2024

Toronto, ON

Software Engineer Intern

Altis Labs

- Developed image annotation tools and data pipelines using **TypeScript**, **GraphQL**, and **Hasura with PostgreSQL** for Altis Labs' AI-powered clinical information platform used by companies such as AstraZeneca and Bayer
- Continued the development of internal **Python** scripts responsible for populating databases with 4000 medical (DICOM) images and over 10 GB of synthetically-generated sample patient data across 148 tables
- Built a comprehensive patient study labeling system by designing database infrastructure and integrating it into the **Hasura GraphQL API**, complete with multiple **Express.js** endpoints for custom input validation on labels' JSON metadata
- Automated the backup of **AWS RDS** databases stored across **S3**, using the **AWS SDK** and **Serverless** functions

Sep. 2023 – Dec. 2023

Toronto, ON

Quality Assurance Intern

Altis Labs

- Improved on the **GitHub Actions CI/CD pipeline** by introducing new frameworks, test reporters, coverage reports, and Python dependency-caching into the project and reduced the runtime of the testing workflow by 68%
- Implemented and configured the tox framework for efficient Python virtual environment management, enabling the ability to run over 250 Python tests with various different **Pipenv environments** collectively
- Wrote over 40 unit and integration tests using **Pytest** and **Jest** to expand test coverage in the code base to over 90%

Jan. 2023 – Apr. 2023

Toronto, ON

PROJECTS

Ticket Finder | *React, Spring Boot, Java, Maven, MySQL, Docker*

- Developed a **Spring Boot** back-end service with a **MySQL** database for updating event details, and a **React** front-end which fetches from both the back-end and the Ticketmaster API, using **Material UI** components to display graphs and data

Live Transcriptions | *JavaScript, Chrome API, OpenAI API*

- Coded a Chrome extension that generates live transcripts from any audio through the web browser regardless of the source, using the **Chrome API** and **OpenAI Whisper API**

Large Language Model | *Python, PyTorch*

- Built a Transformer-based large language model with 124M parameters, utilizing multi-head self-attention mechanisms and pretraining the model on a custom text dataset for 10 epochs

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C/C++, SQL, GraphQL, Bash, HTML, CSS

Libraries/Frameworks: React, Node, Express, Spring Boot, FastAPI, Apollo Client, Serverless, Cypress, Pytest, Jest, PyTorch

Developer Tools: Git, Docker, PostgreSQL, MySQL, AWS, GCP, Terraform, GitHub Actions, Vercel, Hasura